# Does Episodic Memory Help Close the Lexical Frequency Gap in Sensitivity to Syntactic Contrasts? A Test Using Retrieval-Augmented Language Models

## Pretrained models and datasets

**Child-realistic pretraining data.** We constructed a dataset to approximate the language input a child receives during early development, drawing on child-directed speech from CHILDES (MacWhinney, 2000), children's stories, and audiobook transcriptions. The complete dataset comprises 50.03M words, consistent with estimates of cumulative linguistic input by early childhood (Gilkerson et al., 2017; Frank, 2023). Table 1 shows the distribution of constituent sub-datasets.

Table 1: Composition of child-realistic pretraining corpus combining child-directed speech, conversational dialogue, and children's books to approximate naturalistic language exposure during early childhood.

| Dataset | Domain | # words | Prop. (%) |
|---|---|---|---|
| CHILDES | CDS | 15M | 30.0 |
| SWBD | Dialogue | 1.18M | 2.4 |
| CSTC | Books | 3.22M | 6.4 |
| OSub | Dialogue | 31.28M | 61.2 |
| **Total** | | **50.03M** | **100.0** |

**Large-scale pretraining data.** For the large-scale condition, we use GPT-2 XL (1.5B parameters) pretrained on the WebText corpus (Radford et al., 2019), which contains approximately 40GB of text scraped from internet sources.

**Model architectures and training.** Tables 2 and 3 summarize the architectural specifications for both models. The child-realistic model uses the GPT-2 Small architecture with 124M parameters, trained from scratch for 10 epochs, following the BabyLM Challenge setup (Hu et al., 2024). The GPT-2 XL model uses the publicly available pretrained checkpoint from Radford et al. (2019).

Table 2: Model architecture specifications. GPT-2 Small was trained from scratch on child-realistic data (50M words); GPT-2 XL uses the publicly released pretrained checkpoint from Radford et al. (2019) trained on WebText (∼40GB).

| Parameter | GPT-2 Small | GPT-2 XL |
|---|---|---|
| # parameters | 124M | 1.5B |
| Transformer layers | 12 | 48 |
| Hidden size | 768 | 1600 |
| FFN size | 3072 | 6400 |
| Attention heads | 12 | 25 |
| Attention dropout | 0.05 | 0.1 |
| Embedding dimension | 768 | 1600 |
| Vocabulary size | 50257 | 50257 |

Table 3: Training hyperparameters for GPT-2 Small on the child-realistic corpus, trained for 10 epochs using a standard autoregressive language modeling objective with cross-entropy loss.

| Hyperparameter | Value |
|---|---|
| Max sequence length | 1024 |
| Batch size | 64 |
| Training epochs | 10 |
| Learning rate | 0.0001 |
| LR scheduler | inverse sqrt |
| Warmup steps | 10000 |
| Optimizer | Adam |
| Adam-$\beta_1$ | 0.9 |
| Adam-$\beta_2$ | 0.98 |
| Dropout | 0.1 |
| Weight decay | 0.01 |
| Gradient clipping | 1.0 |

## Evaluation set construction

To ensure focused evaluation on syntactic phenomena relevant to our research questions, we selected subsets corresponding to our target phenomena from three evaluation benchmarks: BLiMP (Warstadt et al., 2020), Zorro (Huebner et al., 2021), and BIG-Bench (Srivastava et al., 2023) as discussed below.

**Source benchmarks** **BLiMP.** The Benchmark of Linguistic Minimal Pairs for English (Warstadt et al., 2020) consists of 67 sub-datasets, each containing 1,000 minimal pairs (67,000 pairs total) isolating specific contrasts in syntax, morphology, or semantics. The data is automatically generated according to linguist-crafted grammar templates, with 96.4% aggregate human agreement with the labels.

**Zorro.** Zorro (Huebner et al., 2021) is a grammar test suite designed for models trained on child-directed language, consisting of 13 phenomena with multiple paradigms. Zorro contains approximately 1,000 minimal pairs per paradigm, with lexical items selected to be compatible with the smaller vocabularies of child-trained models.

**BIG-Bench.** The Beyond the Imitation Game Benchmark (Srivastava et al., 2023) includes numerous linguistic evaluation tasks. We selected relevant subject-verb agreement and syntactic construction tasks from this benchmark.

**Instance selection procedure** For each evaluation instance in the original datasets, we selected instances containing one or more of the following syntactic structures using spaCy (Honnibal et al., 2020).

**Subject-Verb detection.** We applied spaCy's depen-

Table 4: Structural similarity between retrieved and target sentences measured by dependency arc overlap (GPT-2 XL, sequence-level, $k = 1$). Values represent the proportion of target dependency relations matched by retrieved sentences.

| | Freq | FastText | GloVe | GPT2-Avg | GPT2-Context |
|---|---|---|---|---|---|
| SV | high | 0.49 | 0.47 | 0.61 | 0.82 |
| | low | 0.45 | 0.42 | 0.58 | 0.78 |
| Wh | high | 0.51 | 0.48 | 0.63 | 0.81 |
| | low | 0.47 | 0.44 | 0.59 | 0.79 |
| RCs | high | 0.50 | 0.49 | 0.62 | 0.80 |
| | low | 0.46 | 0.43 | 0.60 | 0.78 |

dency parser to identify subject-verb relationships. Instances were flagged if they contained `nsubj` (nominal subject) or `nsubjpass` (passive nominal subject) dependencies linking a noun phrase to a finite verb.

**Wh-Sentence detection.** We used part-of-speech (POS) tagging to identify wh-words (POS tags: `WDT`, `WP`, `WP$`, `WRB`) at clause boundaries, following standard Penn Treebank conventions (Marcus et al., 1993). Instances containing interrogative or relative wh-elements were marked for inclusion.

**Relative Clause detection.** We combined POS tagging with dependency parsing to identify relative clause structures. Specifically, we detected: (1) Relative pronouns (`WDT`, `WP`) or relativizers (*that*, *which*, *who*); (2) Clausal complements with `relcl` (relative clause modifier) dependency relations; and (3) Appropriate structural embedding within a noun phrase.

## Structural similarity analysis

Structural similarity (used in the analysis in Section **??**) is defined as dependency arc overlap between retrieved and target sentences. We used spaCy's dependency parser (Honnibal et al., 2020), measuring the proportion of matching subject, object, and modifier relations. Table 4 shows that full contextualized retrieval (GPT2-Context) maintains substantially higher structural similarity than uncontextualized embeddings (FastText, GloVe). Averaged contextualized embeddings (GPT2-Averaged) show intermediate structural preservation, consistent with our interpretation in Section **??** that averaging reduces but does not fully eliminate structural information. These structural similarity scores directly correspond to the performance patterns in Table **??**: high structural similarity correlates with retrieval benefits, while low structural similarity leads to performance at or below baseline.

## Full Retrieval Configuration Results

This section provides complete results across all retrieval configuration combinations discussed in Section **??**. We vary three parameters independently: number of retrieved neighbors ($k \in \{1, 16, 1024\}$), context window size ($\tau \in \{1, 3, 10\}$), and retrieval granularity (token, phrase, sequence). Table 5 reports accuracy for representative configurations ($\tau \in \{3, 10\}$, $k = 16$) across both model scales, andFigures 1–8 provide complete visualizations across all parameter combinations.

**Effects of context window size.** Figures 1–4 show the effect of context window size ($\tau \in \{1, 3, 10\}$) across granularities, phenomena, frequency bands, and both model scales. The most pronounced pattern is that **sequence-level retrieval benefits most from larger context windows in structurally complex phenomena like relative clauses**. For example, in GPT-2 XL at $k = 1$, sequence-level retrieval for low-frequency relative clauses shows clear improvements as $\tau$ increases from 1 to 10, while the corresponding high-frequency items remain largely stable (Figures 1f and 1i). By contrast, there is no consistent pattern on token- and phrase- level across syntactic phenomena.

**Effects of number of retrieved neighbors.** Figures 5–8 show the effect of varying $k \in \{1, 16, 1024\}$ across granularities, phenomena, and frequency bands. The non-monotonic pattern reported in the main text is robust across all configurations: moving from $k = 1$ to $k = 16$ consistently improves performance for low-frequency items, while increasing further to $k = 1024$ leads to degradation. This pattern is most pronounced for sequence-level retrieval and structurally complex phenomena: for both GPT-2 XL and the child-realistic model, at $\tau = 3$, sequence-level retrieval for low-frequency RCs peaks at $k = 16$ compared to $k = 1$ and $k = 1024$. The degradation at $k = 1024$ is most pronounced for sequence-level retrieval and for structurally complex phenomena. For simpler phenomena such as subject-verb agreement, and for high-frequency items more generally, both the gains at $k = 16$ and the degradation at $k = 1024$ are more modest.

Comparing $\tau = 1$(Figures 5 and 7) against $\tau = 3$ (Figures 6 and 8) shows that larger context windows amplify the benefit of moderate aggregation: at $\tau = 3$, moderate aggregation at $k = 16$ provides stronger gains for sequence-level low-frequency items compared to $\tau = 1$. The degradation at $k = 1024$, however, remains close at $\tau = 3$ as at $\tau = 1$, confirming that over-retrieval noise cannot be mitigated by wider context alone. The child-realistic model exhibits the same non-monotonic pattern as GPT-2 XL but with a more pronounced drop at $k = 1024$, particularly for low-frequency relative clauses and Wh-questions

## References


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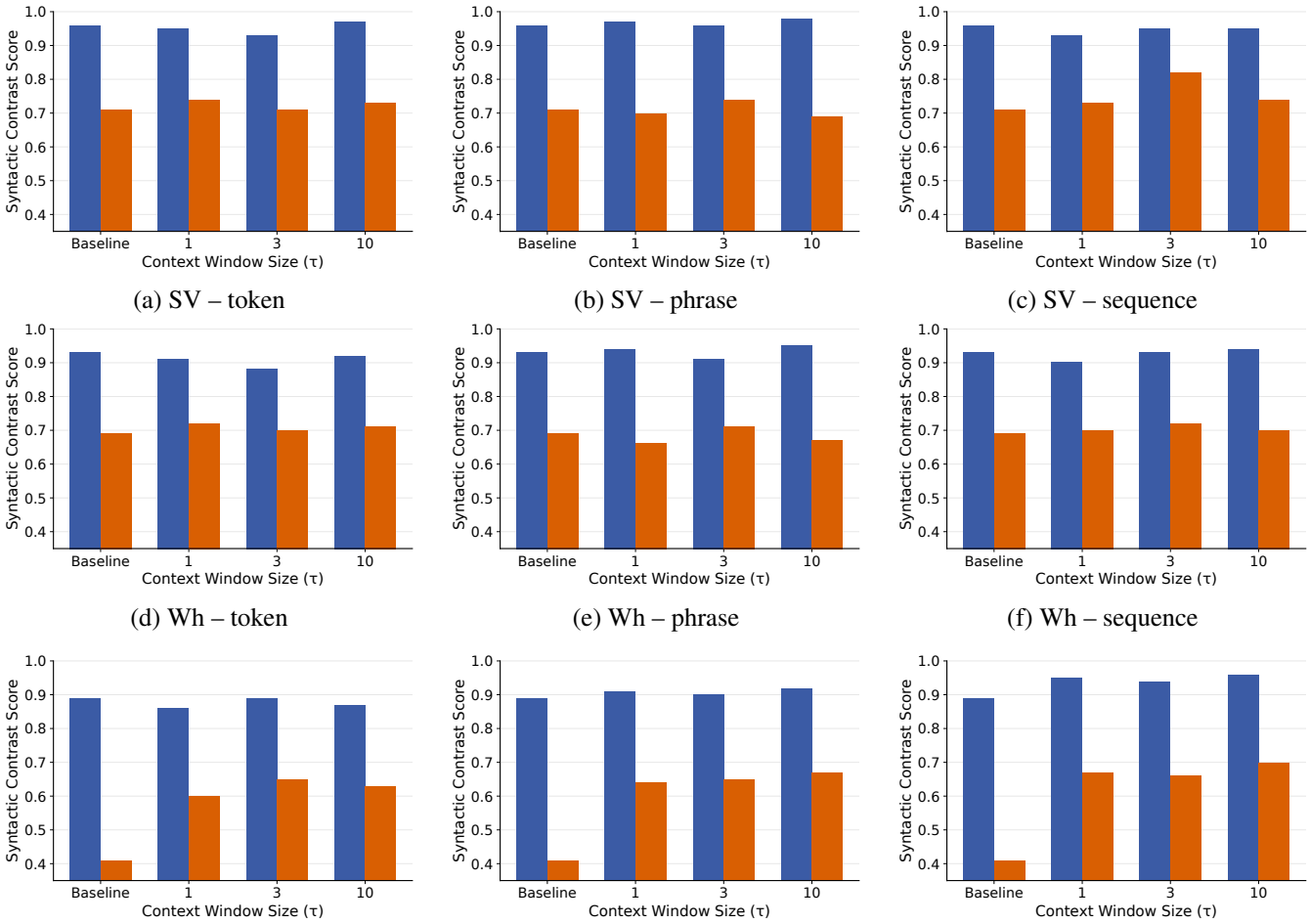
Table 5: Performance by retrieval granularity at $\tau \in \{3, 10\}$ with $k = 16$ for child-realistic and GPT-2 XL models. Sequence-level retrieval consistently yields the strongest gains for low-frequency items in wh-questions and relative clauses across both model scales and context sizes.

| | | | **Child-realistic** | | | | **GPT-2 XL** | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| $\tau$ | Phen. | Freq | baseline | token | phrase | seq. | baseline | token | phrase | seq. |
| 3 | SV | high | 0.91 | 0.91 | 0.90 | 0.90 | 0.96 | 0.95 | 0.95 | 0.95 |
| | | low | 0.60 | 0.76 | 0.74 | 0.76 | 0.71 | 0.82 | 0.81 | 0.83 |
| | Wh | high | 0.92 | 0.89 | 0.88 | 0.88 | 0.93 | 0.93 | 0.93 | 0.93 |
| | | low | 0.61 | 0.67 | 0.69 | 0.70 | 0.69 | 0.72 | 0.73 | 0.74 |
| | RC | high | 0.88 | 0.89 | 0.88 | 0.89 | 0.89 | 0.94 | 0.94 | 0.95 |
| | | low | 0.48 | 0.60 | 0.62 | 0.65 | 0.41 | 0.66 | 0.69 | 0.70 |
| 10 | SV | high | 0.91 | 0.91 | 0.91 | 0.91 | 0.96 | 0.96 | 0.96 | 0.96 |
| | | low | 0.60 | 0.67 | 0.71 | 0.71 | 0.71 | 0.71 | 0.74 | 0.75 |
| | Wh | high | 0.92 | 0.89 | 0.89 | 0.89 | 0.93 | 0.93 | 0.93 | 0.93 |
| | | low | 0.61 | 0.65 | 0.69 | 0.69 | 0.69 | 0.72 | 0.73 | 0.73 |
| | RC | high | 0.88 | 0.84 | 0.86 | 0.85 | 0.89 | 0.89 | 0.90 | 0.90 |
| | | low | 0.48 | 0.37 | 0.65 | 0.68 | 0.41 | 0.41 | 0.68 | 0.71 |

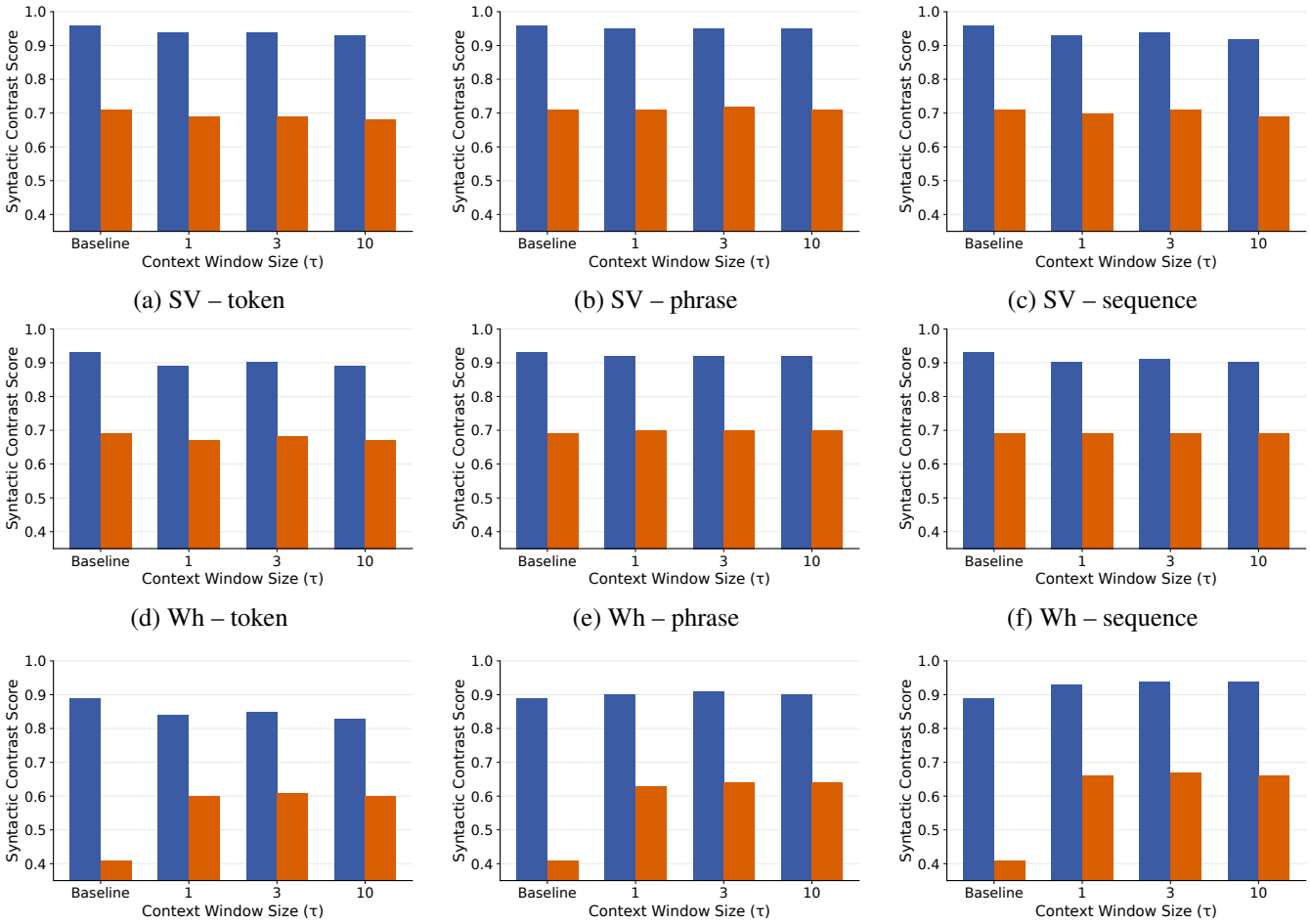



(a) SV – token (b) SV – phrase (c) SV – sequence

(d) Wh – token (e) Wh – phrase (f) Wh – sequence

(g) RC – token (h) RC – phrase (i) RC – sequence

Figure 1: Effect of context window size ($\tau \in \{1, 3, 10\}$) for GPT-2 XL at $k = 1$. Larger context windows benefit low-frequency items more than high-frequency items, particularly for sequence-level retrieval and structurally complex phenomena (Wh, RCs). Subject-verb agreement is less sensitive to context size.

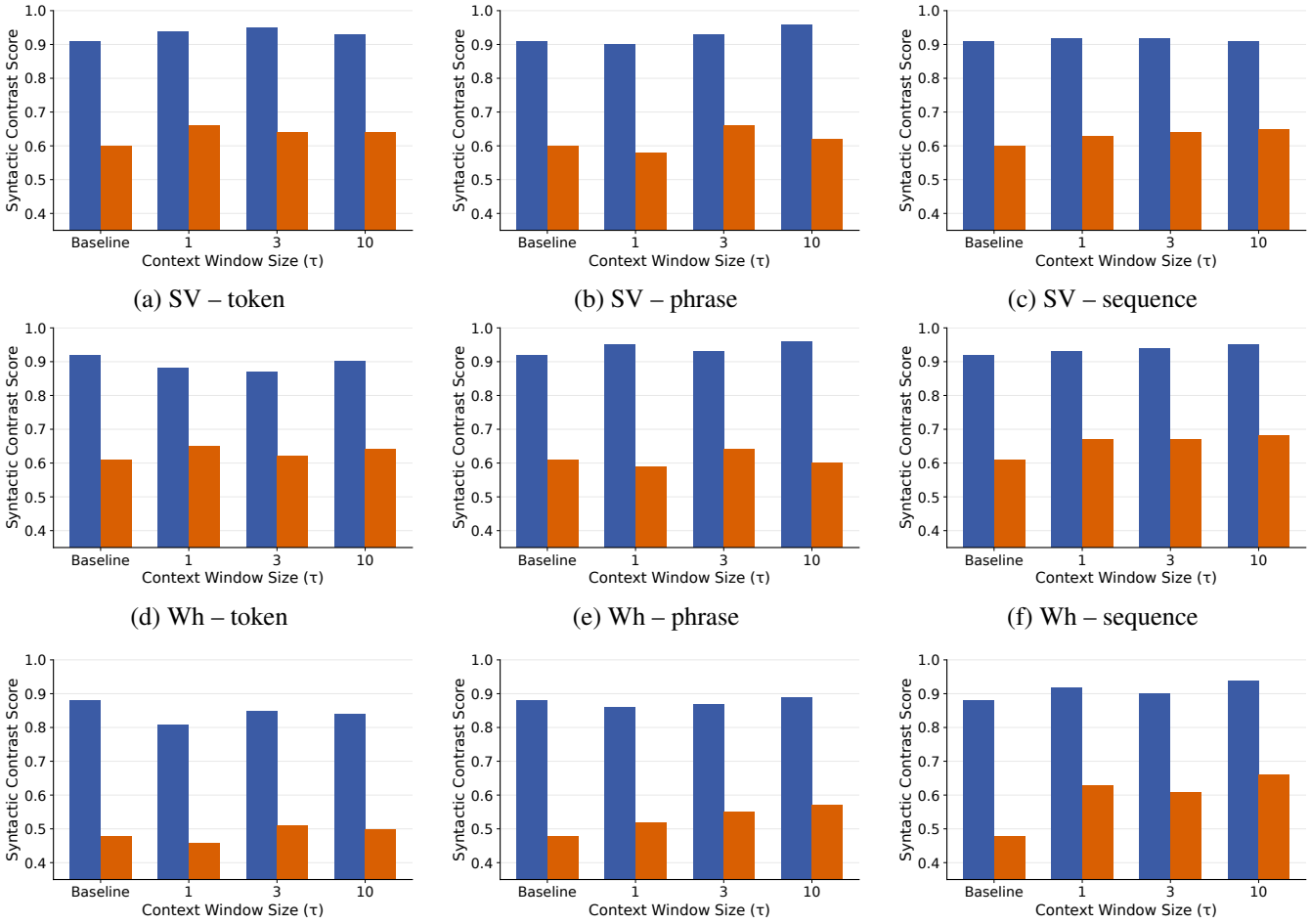



(a) SV – token (b) SV – phrase (c) SV – sequence

(d) Wh – token (e) Wh – phrase (f) Wh – sequence

(g) RC – token (h) RCs – phrase (i) RC – sequence

Figure 2: Effect of context window size ($\tau \in \{1, 3, 10\}$) for GPT-2 XL at $k = 1024$. Compared to $k = 1$, performance degrades substantially for low-frequency items, most severely for sequence-level retrieval, confirming that over-retrieval introduces noise that context window size cannot compensate for.

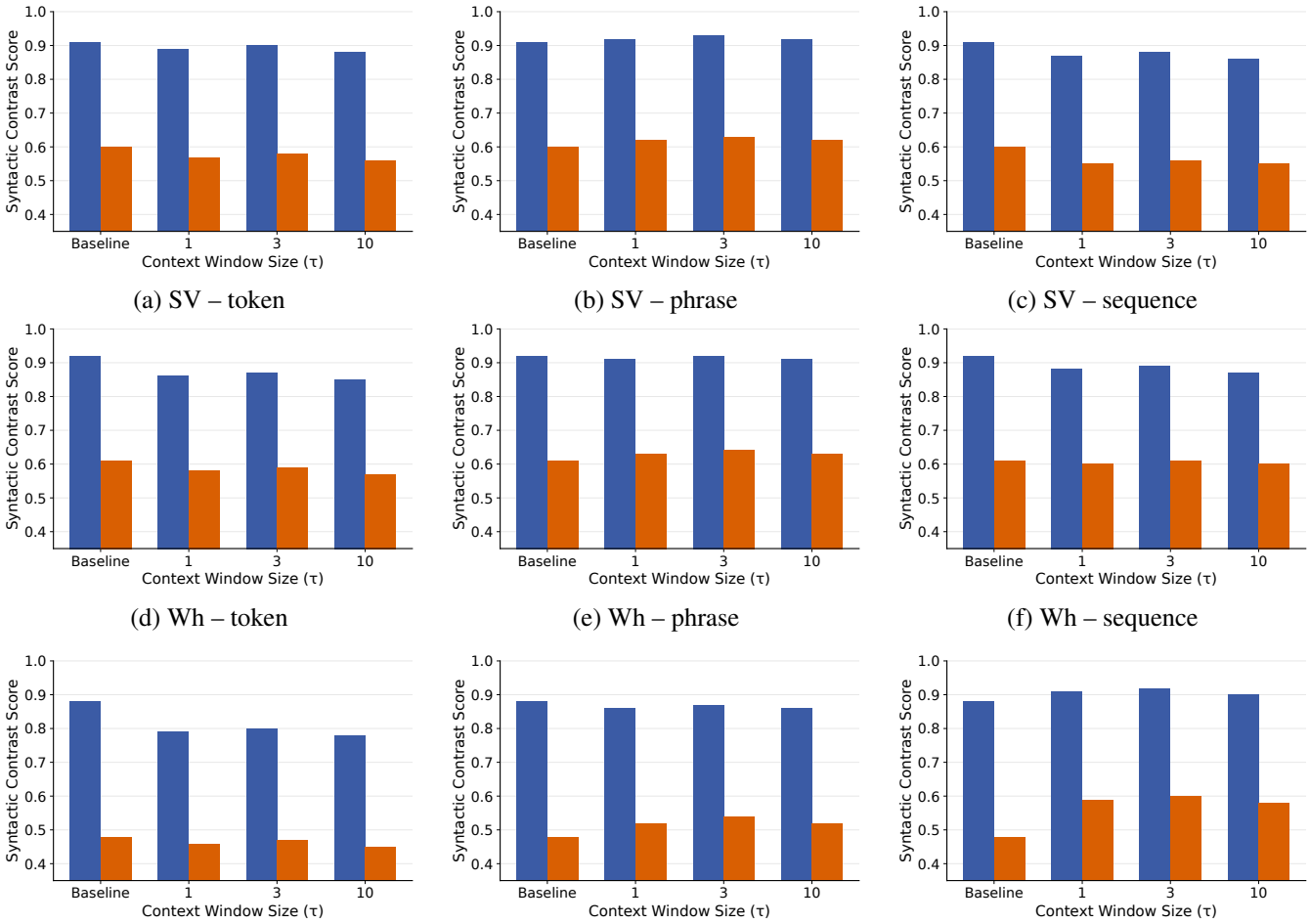



(a) SV – token (b) SV – phrase (c) SV – sequence

(d) Wh – token (e) Wh – phrase (f) Wh – sequence

(g) RC – token (h) RC – phrase (i) RC – sequence

Figure 3: Effect of context window size ($\tau \in \{1, 3, 10\}$) for the child-realistic model at $k = 1$. Patterns are qualitatively similar to GPT-2 XL (Figure 1) but with smaller absolute gains, consistent with the smaller model's weaker parametric representations.

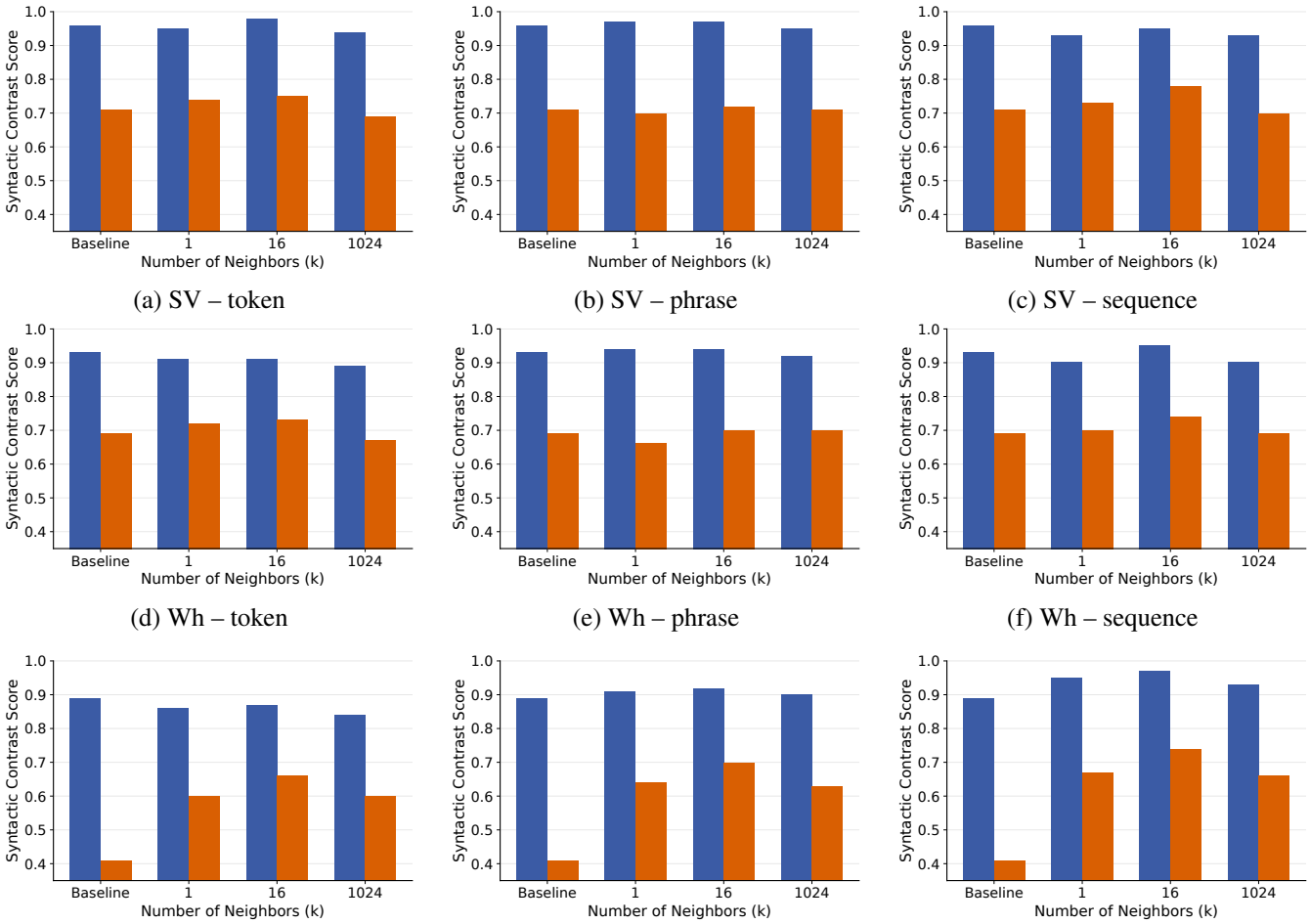



(a) SV – token (b) SV – phrase (c) SV – sequence

(d) Wh – token (e) Wh – phrase (f) Wh – sequence

(g) RC – token (h) RC – phrase (i) RC – sequence

Figure 4: Effect of context window size ($\tau \in \{1, 3, 10\}$) for the child-realistic model at $k = 1024$. Performance degradation at large $k$ is more pronounced than for GPT-2 XL, particularly for low-frequency RCs, suggesting the smaller model is more susceptible to retrieval noise.

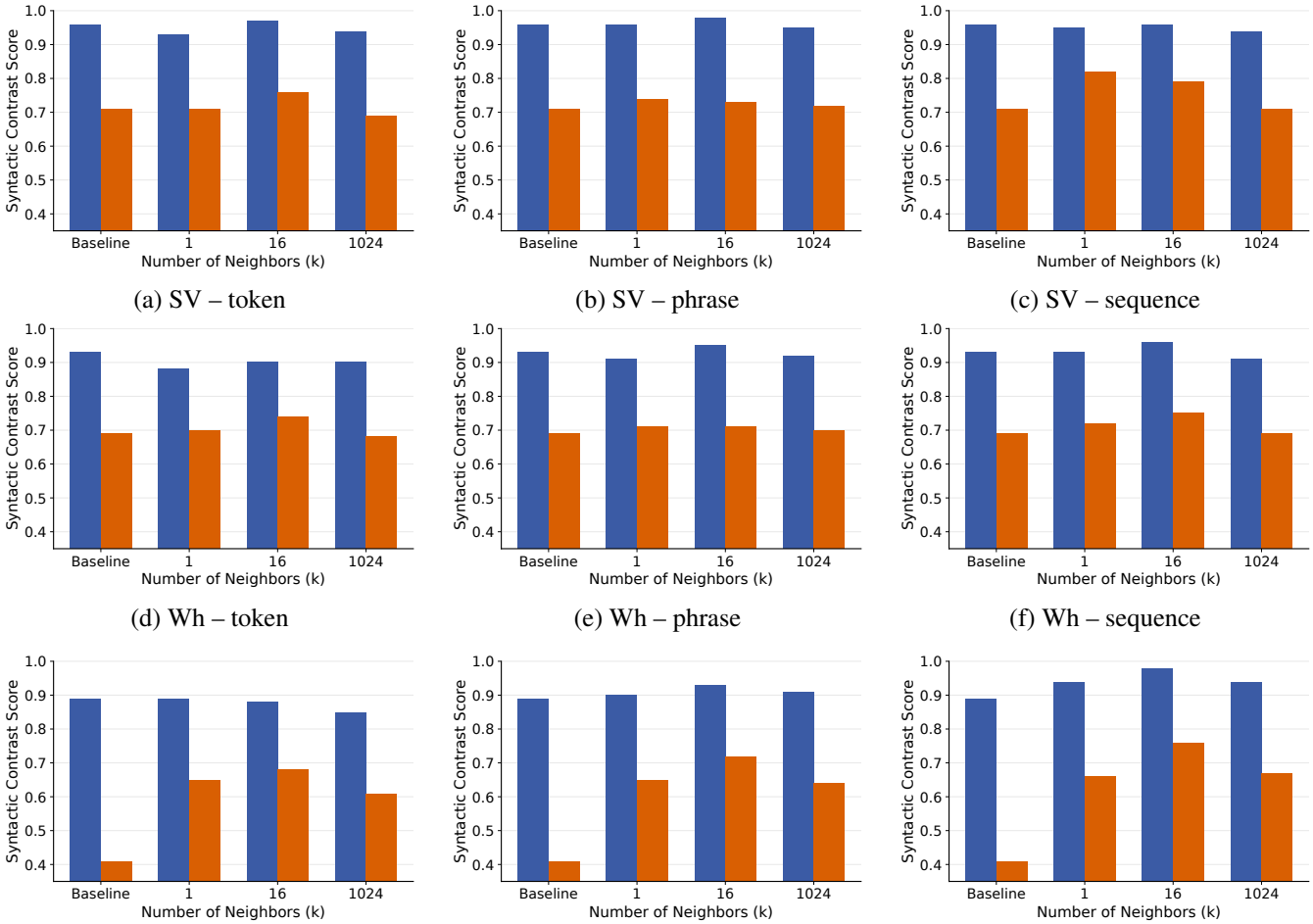



(a) SV – token (b) SV – phrase (c) SV – sequence

(d) Wh – token (e) Wh – phrase (f) Wh – sequence

(g) RC – token (h) RC – phrase (i) RC – sequence

Figure 5: Effect of number of retrieved neighbors ($k \in \{1, 16, 1024\}$) for GPT-2 XL at $\tau = 1$. The non-monotonic pattern holds: $k = 16$ outperforms both $k = 1$ and $k = 1024$ across most conditions, with degradation at large $k$ most severe for sequence-level retrieval and low-frequency items.

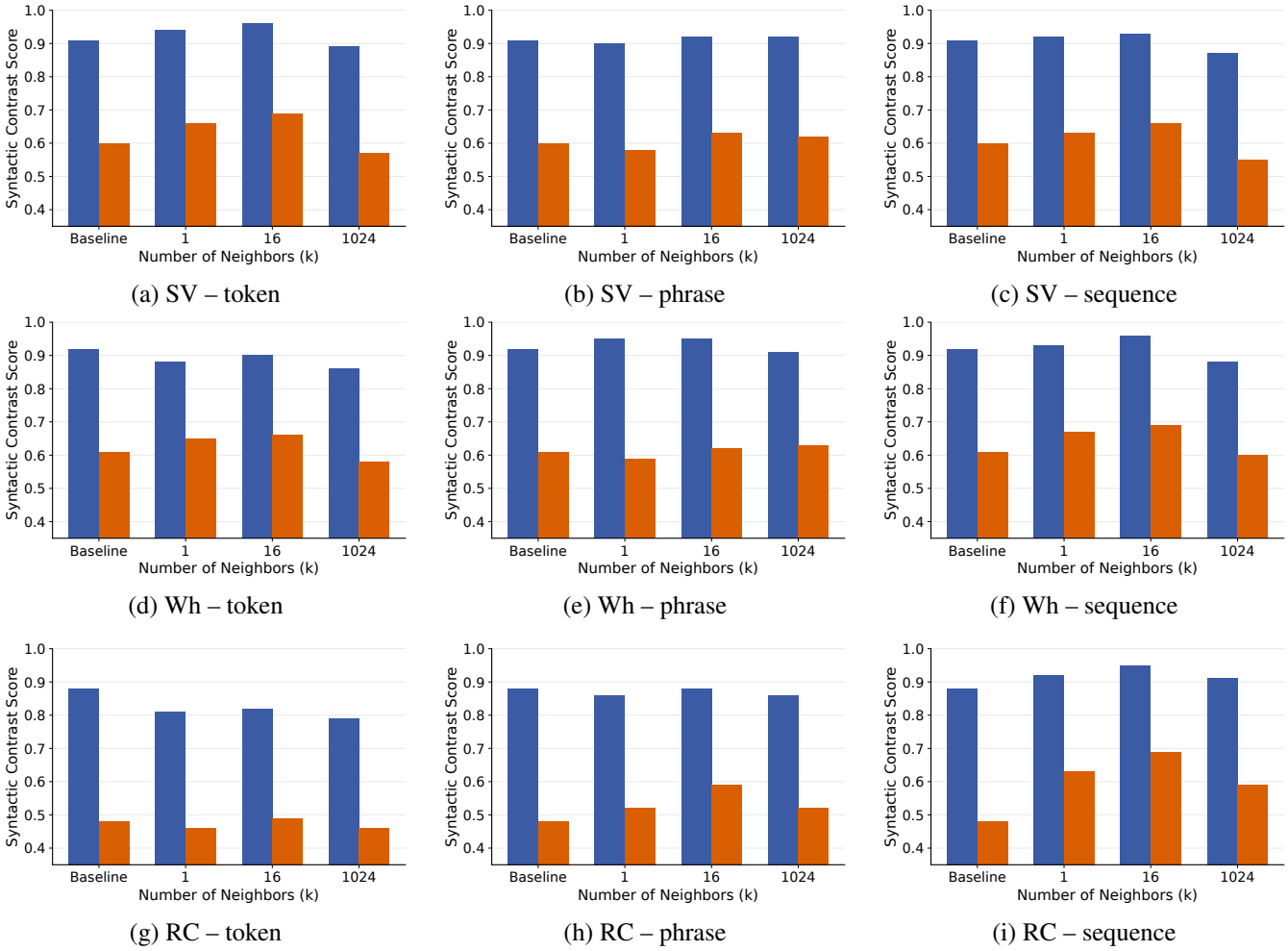



(a) SV – token (b) SV – phrase (c) SV – sequence

(d) Wh – token (e) Wh – phrase (f) Wh – sequence

(g) RC – token (h) RC – phrase (i) RC – sequence

Figure 6: Effect of number of retrieved neighbors ($k \in \{1, 16, 1024\}$) for GPT-2 XL at $\tau = 3$. Results are consistent with $\tau = 1$ (Figure 5): $k = 16$ provides the best balance between evidence aggregation and noise across phenomena and frequency bands.




(a) SV – token (b) SV – phrase (c) SV – sequence

(d) Wh – token (e) Wh – phrase (f) Wh – sequence

(g) RC – token (h) RC – phrase (i) RC – sequence

Figure 7: Effect of number of retrieved neighbors ($k \in \{1, 16, 1024\}$) for the child-realistic model at $\tau = 1$. The non-monotonic pattern holds across both model scales, with $k = 16$ providing the best overall performance and $k = 1024$ degrading most severely for low-frequency RCs.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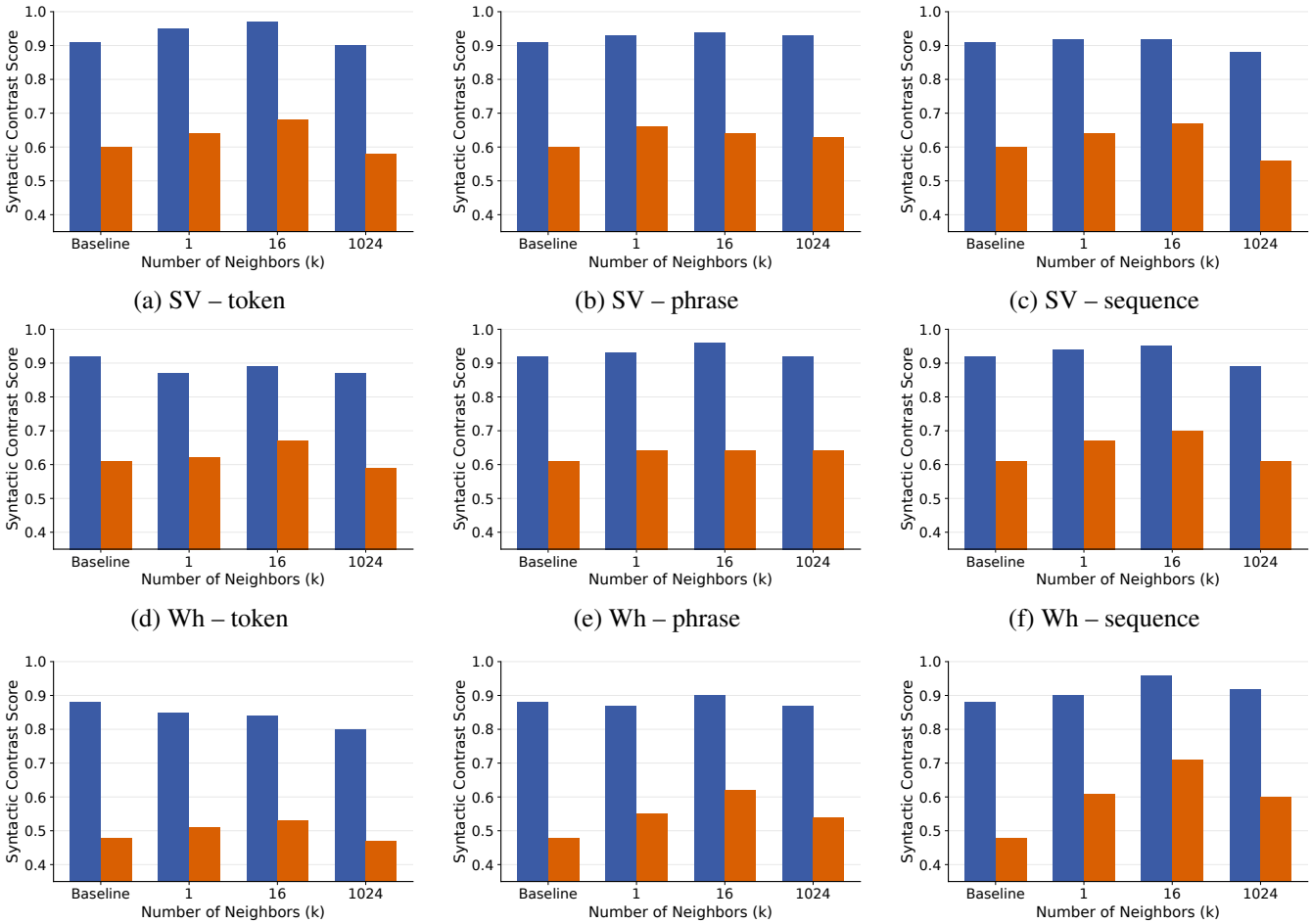


(a) SV – token (b) SV – phrase (c) SV – sequence

(d) Wh – token (e) Wh – phrase (f) Wh – sequence

(g) RC – token (h) RC – phrase (i) RC – sequence

Figure 8: Effect of number of retrieved neighbors ($k \in \{1, 16, 1024\}$) for the child-realistic model at $\tau = 3$. Consistent with all other configurations, $k = 16$ provides the best balance between evidence aggregation and noise, with gains most pronounced for low-frequency RCs and Wh-questions.